Estimating the prevalence of the Sonority Sequencing Principle accross languages and language families

Introduction

Speech is made of hierarchically organised, discrete linguistic units (e.g., phonemes, syllables, words, phrases). Syllables are a privileged linguistic unit in speech processing, particularly during early language acquisition: newborns preferentially chunk the speech signal into syllable-sized units [1], [2]. This is remarkable, given they have accumulated a relatively limited linguistic experience. The mechanisms underlying syllable identification are still unclear.

There is some evidence that, at birth, humans are sensitivity to certain constraints to syllabic structure. One of them is the Sonority Sequencing Principle (SSP). The SSP constrains the order of the segments inside a syllable based on sonority, a phonological feature that describes the perceptual loudness of a segment relative to neighbouring segments. The SSP states that each syllable displays a unique peak of sonority in the nucleus, and that sonority rises unidirectionally from the syllable margins toward the nucleus [3].

Although the specific acoustic correlates of sonority are still under debate [3], [4], sonority scales have been developed and agreed one by most researchers so that different segment classes are assigned some scalar, ordered sonority score, following available literature of cross-linguistic phonology [5, for an acoustic grouding of this sonority hierarchy, based on intensity]. By assigning numeric sonority values to specific segments, syllables can be classified as following or violating the SSP. G. N. Clements [6] proposes the following sonority hierarchy:

vowels > glides > liquids > nasals > obstruents

D. Gomez *et al.* [7] showed in newborns that hearing SSP-following syllables (e.g., *bran*) elicits different brain haemodynamic responses compared to SSP-violating syllables (e.g., *lbif*), suggesting that newborns, how have not yet acquired extensive experience with the phonological contraints of their native language, are already sensitive to violations of the SSP. Such early bias toward certain syllabic structures would provide humans with an available mechanism to detect syllables in the continuous speech stream at the earliest stages of language acquisition.

The SSP has been long claimed to be a linguistic universal, on the grounds of SSP-following syllables being preferred over SSP-violating syllables across most languages. S. Parker [8] collects evidence from 275 experimental works on a typologically diverse list of 75 distinct languages, and reports that 63% of the studies suggest that sonority plays a relevant role for accounting for the results, and 22% provide inconclusive evidence, and 15% provide negative evidence for such a relevant role of sonority.

The universality of the SSP would place this principle as one of the few cues for speech segmentation present from birth to *all* humans, before language-specific constraints to sound sequences (e.g., phonotactics) are available. The universal status of the SSP is however a matter of debate, as violations of the SSP are common across languages [9]. For instance, the /st-, -ts/ consonant clusters violate the SSP when at syllabic onset or at coda (e.g., *street*, *ticks*), respectively. This is the case across many languages [10]. Some languages, like Russian, are known for allowing fairly complex consonant clusters at syllabic onset that violate the SSP (e.g., взгляд (*vzglyad* [glance]) [11], [12].

More recent accounts of linguistic universals [13] propose a softer understanding of the term "universal", such that linguistic universals represent overall preferences toward certain linguistic structures across languages, instead of hard ON/OFF constraints. Under this account, the role of the SSP as a universally available cue for syllabification becomes a statistical question: how often do syllables in different languages abide by the SSP? In this line, S. A. Frisch [14] estimates the relative frequency of SSP-following consonant clusters across languages 93 langugaes at around 70%, compared to SSP-violating clusters.

The aim of the present paper is to provide empirical estimates of the prevalence of the SSP in a fairly representative sample of languages. We pursued this goal in two ways. In a first analysis, we estimated the proportion of the syllabic repertoire of each language that followed SSP. Because items inside the syllabic repertoire of a language likely occur at different rates in speech (e.g., both /ka/ and /gla/ syllables are attested in Spanish, but the former appears in a larger number of words), a second analysis investigated whether SSP-following syllables are also more frequent speech compared to SSP-violating syllables. We explored this research question in two datasets. In the first dataset, syllabic frequencies were derived from available databases of lexical frequencies that did not discriminate between adult-directed speech (ADS) and child-directed speech (CDS). In the second dataset, syllabic frecuencies were derived from lexical frequencies in the CHILDES [15], which includes corpora of child-directed speech.

Analysis 1: What proportion of the syllabic repertoire of a language follows the SSP?

Methods

Data collection

We scraped word entries with phonological transcriptions from Wiktionary. Wiktionary is an open-source, free database of lexical entries. As of June 10th 2026, Wiktionary contains 10,050,751 lexical entries from 8,244 langages contributed by 4,463,976 total users. Some lexical entries in Wiktionary also include a phonological in IPA format or phonetic transcription of the word-form in a particular language. We used the `wikipron` Python package [16] to scrape all available phonological transcriptions from all 7859 languages included in the Glottolog database [17]. Using the `lingtypology` Python package [18], [19], we extracted their corresponding ISO-639-3 code and accompanying information (e.g., language family, official name). This resulted in ~1.7 million pronunciations belonging to 900 languages. We filtered out languages with less than

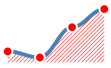
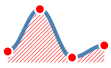
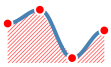
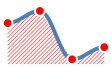
100 phonological transcriptions available. The final dataset containing word-form phonological transcriptions included 731,145 word-forms from 192 languages.

Data processing

We segmented word-form phonological transcriptions into syllabic candidates where possible. If available, we used explicit syllable boundary markers (`.`) or implicity syllable markers (`ˈ` for primary stress, `ˌ` for secondary stress). In the absence of syllabic markers, we assumed the word to be monosyllabic. We then classified the segments inside each resulting syllable candidate into consonants (`C`), vowels (`V`), and semi-vowels (`v`). Consecutive sequencies of vowels and semi-vowels (e.g., `V`, `VV`, `Vv`, `VVv`) where classified as vocalic clusters. Syllables with more than one vocalic cluster were flagged as multisyllabic, and were excluded from the dataset. We also remove duplicated syllables within each language (e.g., if the syllable /ka/ appeared in Spanish words /ka.sa/ and /kaˈpo/ in Spanish, it would appear more than once in the Spanish dataset), resulting in a syllabic repertoire for each language. We then assigned a sonority score to each segment in each syllable in a scale from 1 to 9, according to the `panphon` Python package [20]. Finally, we evaluated whether each syllable followed or violated the SSP. Syllables with more than one peak (i.e., local minimum) in the sonority sequence were classified as SSP-violating, and as SSP-following otherwise. Syllables with a flat sonority contour were classifid as SSP-following.

We then used the `wikipron` Python package [16] to scrape the Wiktionary page of each language looking for pronunciations. First, narrow phonological transcriptions were searched. If no narrow phonological transcriptions are available, broad phonological transcriptions were searched. Finally, if no broad phonological transcriptions are available either, the language was skipped. We then filtered our languages with less than 100 available pronunciations. This resulted in 7859 languages with at least 100 pronunciations available in Wiktionary.

Table 1: Examples of SSP-following and SSP-violating syllables in the dataset. Sonority index contours are shown for each syllable numerically and graphically.

Language	Syllable	Original word	Sonority index	
SSP-following				
Kazakh	/bas/	қотақбас /qo.taq.bas/	2-9-3	
Mon	/baŋ/	ကွင်ကျာ /ka.baŋ.kjaː/	2-9-5	
Polish	/alt/	Romuald /rɔˈmu.alt/	9-6-1	
Sanskrit	/bim/	प्रतिबिम्ब /prɐ.t̪i.bim.bɐ/	2-8-5	
Tajik	/ast/	аст /ast/	9-3-1	
SSP-violating				
Tibetan	/bkra/	སྦ་བཀྲ /*sa.bkra/	2-1-6-9	
Moselle Franconian	/bats/	Batz /bats/	2-9-1-3	
Polish	/atr/	teatr /ˈte.atr/	9-1-6	
Tajik	/jabr/	ноябр /noˈjabr/	7-9-2-6	
Uighur	/ʔɛks/	ئەكس /ʔɛks/	7-9-1-3	

Data analysis

To model the prevalence of the SSP across languages, we aggregated, for each language in the dataset, the number of syllables that followed the SSP. This resulted in one data point per language. We used a multilevel, Bayesian logistic regression model to estimate the proportion of syllables that followed or violated the SSP across languages. This model estimated the average probability of the syllables of language family following the SSP. This model included each family language as a group with a random intercept. Equation 1 shows a detailed implementation of both

models model. We implemented the model in the probabilistic programming language Stan [21] and CmdStanPy, its interface to its Python (3.12.7). We run eight Hamiltonian Monte Carlo chain of 2,000 iterations each (1,000 were warm-up iterations) to estimate the posterior distribution of the model.

Because we did not have access to the full syllabic repertoire of each language, available syllables from languages are the result of convenience sampling (e.g., in Wiktionary word entries are created by volunteers). The number of entries from each language, the amount of detail with which they are entered, and the type of words that are entered, varies systematically across languages. This might bias the estimates. For instance, if SSP-violating syllables are less frequent than SSP-following syllables in the lexicon of a language, and less frequent words are less likely to enter Wiktionary by volunteers, then the observed prevalence o fSSP for that language likely overestimates the prevalence of SSP-following languages. For this reason, this model included a term in the linear model γ/n_i aimed at capturing this potential overrepresentation of SSP-following syllables in datasets with fewer observations.

Likelihood:

$$
\begin{aligned}
y_j &\sim \text{Binomial } (n_j, \frac{\theta_j}{1-\theta_j}) \\
\theta_j &= \mu_j + u_j + (\gamma/n_j)
\end{aligned}
$$

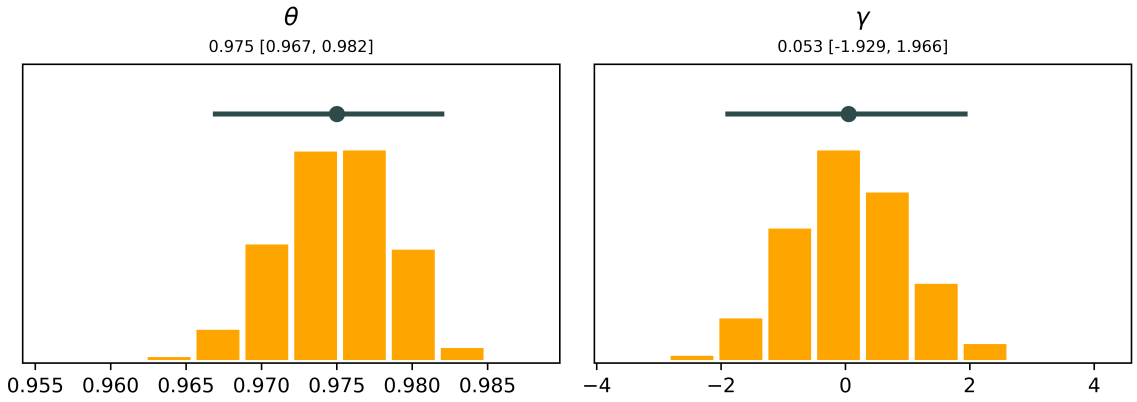
where:

$$
\begin{aligned}
&i \text{ represents each observation, corresponding to a language} \\
&y_j \text{ is the number of syllables in language } i \text{ that follow SSP} \\
&u_j \text{ is a language-wise adjustment} \\
&n_i \text{ is the number of syllables total language } i \\
&\gamma \text{ introduces an adjustment to } \mu \text{ that is inverselly proportional to } n_i
\end{aligned}
\tag{1}
$$

Prior:

$$
\begin{aligned}
\mu_j &\sim \text{Normal } (0, 5) \\
\sigma_u &\sim \text{Exponential } (4)
\end{aligned}
$$

Results


θ
0.975 [0.967, 0.982]
0.955
0.960
0.965
0.970
0.975
0.980
0.985
γ
0.053 [-1.929, 1.966]
−4
−2
0
2
4

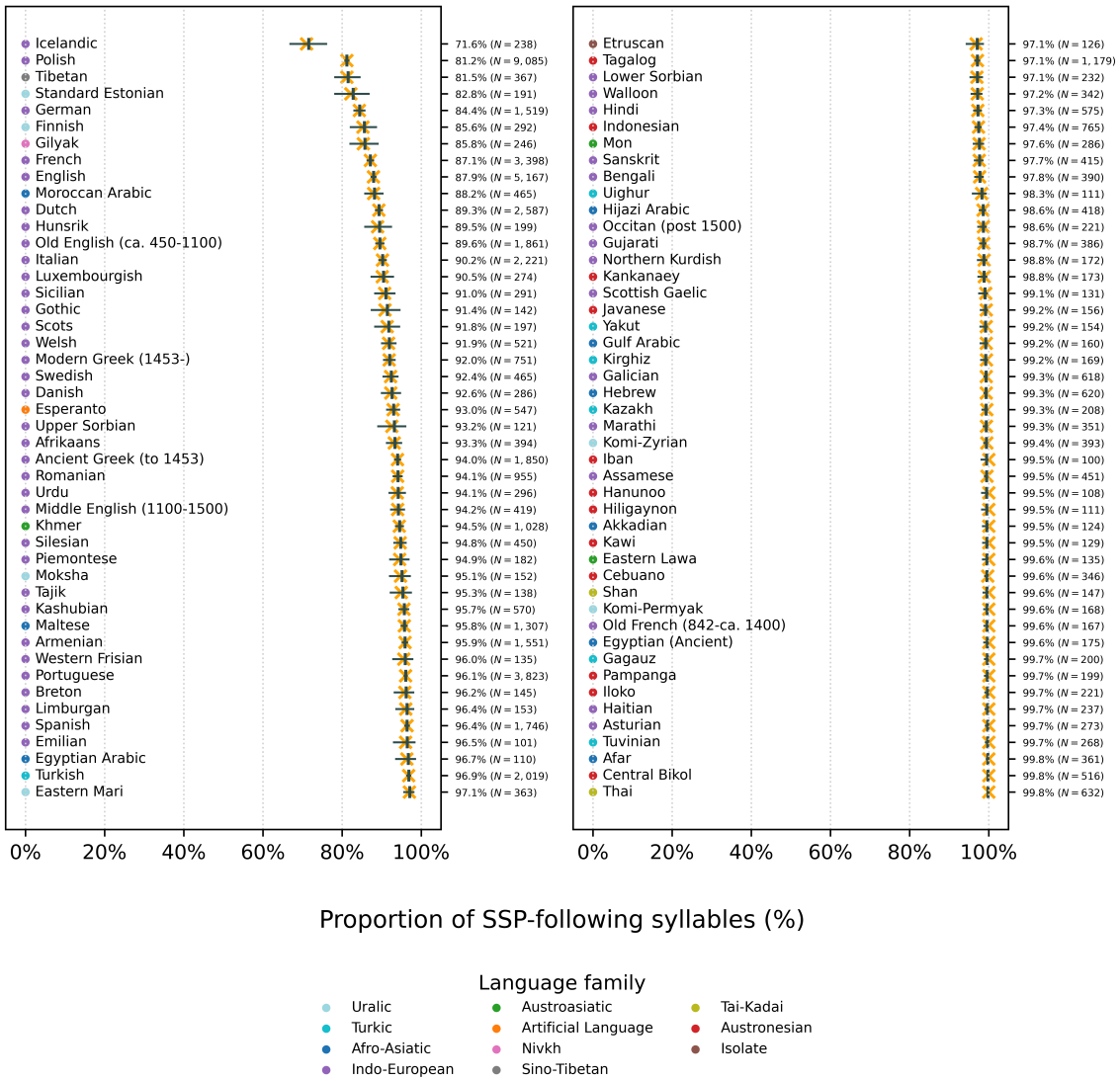


Figure 1: Posterior model-estimated proportion of distinct syllables that follow the Sonority Sequencing Principle (SSP) for each language family. The X-axis indicates the proportion of syllables that follow the SSP. The Y-axis indicates each language family. Black diamonds and intervals indicate the posterior mean proportion of SSP-following syllables for each language family, and its 95% highest density interval (HDI). Means and 95% HDI are also numerically reported next to each language family.

Analysis 2: Are SSP-following syllables more frequent than SSP-violating syllables?

1) Create a *syllabic frequency* measure that accounts for how often a syllable appears across words in each language, and that adjusts this prevalence by the lexical frequency of the words it appears in [14, for a similar approach]. 1.1) For each language dataset, get the lexical frequency

of each word in a lexical frequency database for that language. 1.2) *Syllabic frequency*: Add up the lexical frequency of the words each syllable appears in.

2) Fit a model (same basic stucture as before) that includes the *syllabic frequency* predictor, and compare its regression coefficient against zero (perhaps specifying a ROPE).

Method

Data collection

We extracted adult lexical frequencies for each word-form in the words dataset using the `wordfreq` Python package. We then performed the same steps as described above to expand the dataset into syllables, and evaluate if each of such syllables followed or violated the SSP. In addition, we computed a measure of syllabic frequency: for each syllable, we aggregated the lexical frequency of each word the syllable appeared in. The rationale behind this step is that each time a word is uttered, each of its syllables is also uttered. Therefore, by calculating the lexical frequency of each word a given syllable is involved in, an estimation of syllabic frequency can be derived.

Data analysis

$$
\begin{aligned}
&\textbf{Likelihood}: \\
&\quad y_{ij} \sim \text{LogNormal } (\mu_{ij}, \sigma_{ij}) \\
&\quad \mu_{ij} = \beta_0 + u_{0j} + (\beta_1 + u_{1j}) \cdot \text{Frequency}_{ij} \\
&\quad \sigma_{ij} = \exp(\pi_0 + w_{0j} + (\pi_1 + w_{1j}) \cdot \text{Frequency}_{ij}) \\
&\textbf{Prior}: \\
&\quad \beta_{[0,1]} \sim \text{Normal } (0, 10) \\
&\quad \pi_{[0,1]} \sim \text{Normal } (0, 5) \\
&\quad u_{[0,1]j} \sim \text{Normal } (0, \tau_{uj}) \\
&\quad w_{[0,1]j} \sim \text{Normal } (0, \tau_{wj}) \\
&\quad \tau_{uj}, \tau_{wj} \sim \text{Exponential } (2)
\end{aligned}
\tag{2}
$$

Results

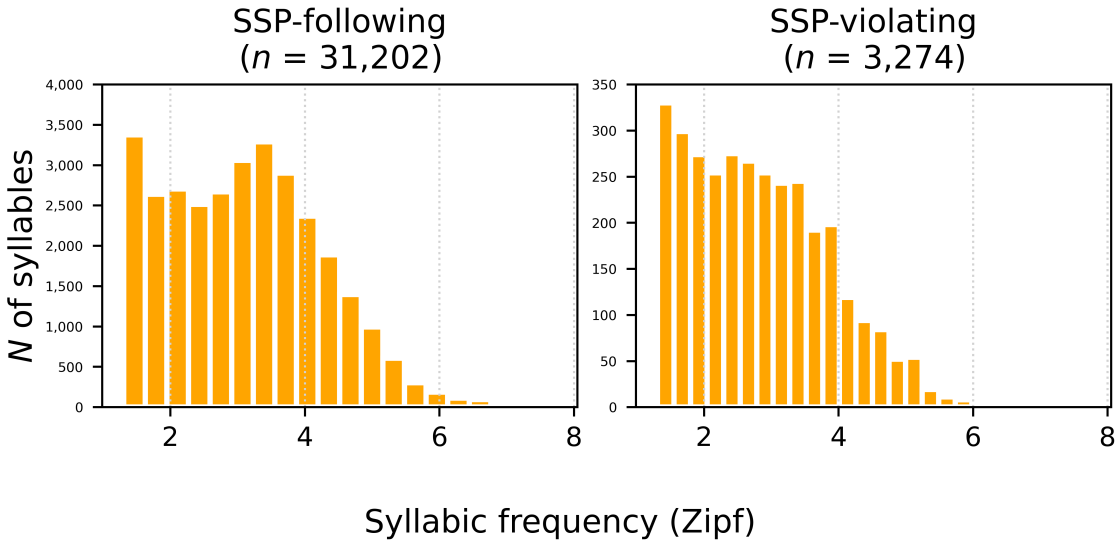
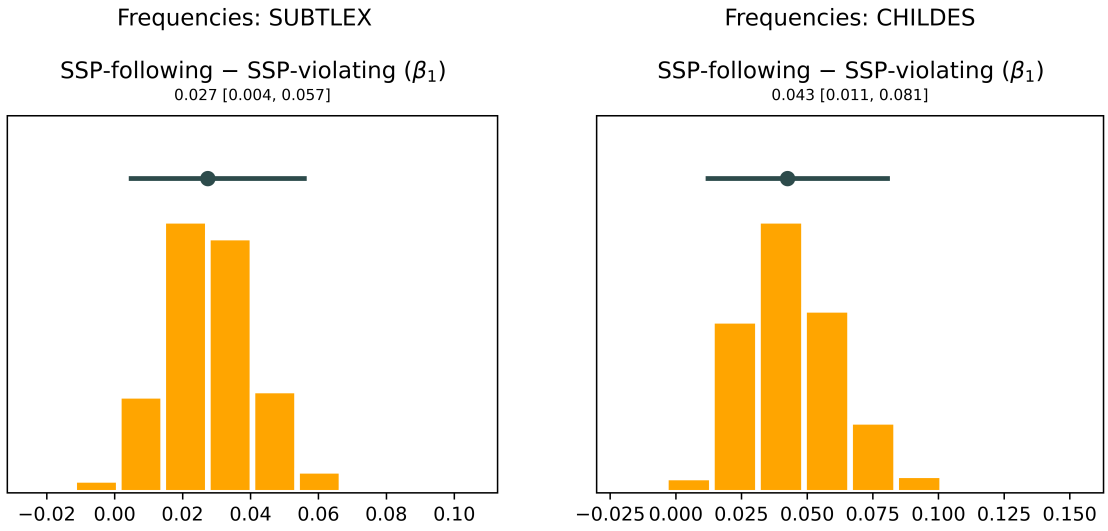


Figure 2: Distribution of syllabic frequencies across languages for SSP-followingg and SSP-violating syllables.



(a) Adult lexical frequencies (b) Infant lexical frequencies (CHILDES)

Figure 3: Posterior distribution of main parameters in Equation 2.

Appendix

Appendix

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